

OIS-004 — Event Implementation

Status: Foundational

Authority: Implementation

Depends on:

- OIS-000 through OIS-003
 - OAS-010 — Event Architecture
 - OCS-000 through OCS-032
-

Purpose

This specification defines the implementation responsibilities for Constitutional Events.

The Architecture defines the role of Events.

This specification defines how executable systems realize immutable constitutional history.

Implementation SHALL preserve Events.

Implementation SHALL derive operational state from Events.

Implementation SHALL never rewrite constitutional history.

1. Implementation Objective

The objective of Event Implementation is to guarantee that every constitutionally significant change is permanently represented as an immutable operational Event.

Implementation SHALL preserve:

Identity.

Ordering.

Context.

Relationships.

Historical continuity.

Recoverability.

Events SHALL become the operational foundation of constitutional continuity.

2. Event Recording

Every constitutionally significant change SHALL generate one or more Events.

Every Event SHALL preserve:

Canonical identity.

Constitutional time.

Origin.

Context.

Affected constitutional constructs.

Applicable implementation metadata.

Implementation metadata SHALL remain operational.

It SHALL never redefine constitutional meaning.

3. Event Immutability

Recorded Events SHALL remain immutable.

Implementation SHALL NOT:

Modify historical Events.

Delete historical Events.

Replace historical Events.

Corrections SHALL generate new Events.

Historical continuity SHALL remain observable.

4. Event Ordering

Implementation SHALL preserve constitutional ordering independently of physical execution order.

Ordering SHALL remain reconstructable across:

Parallel execution.

Distributed execution.

Federation.

Recovery.

Migration.

5. Event Streams

Every constitutional construct SHALL expose an operational Event Stream.

Streams SHALL preserve:

Creation.

Revision.

Migration.

Federation.

Participation.

Lifecycle evolution.

Streams SHALL remain independently reconstructable.

6. Derived State

Operational state SHALL be treated as a projection derived from Event history.

Implementations MAY:

Cache projections.

Materialize projections.

Optimize projections.

Derived state SHALL remain disposable.

Events SHALL remain authoritative.

7. Event Storage

Implementations MAY employ:

Relational databases.

Document databases.

Append-only logs.

Distributed ledgers.

Object stores.

Storage SHALL preserve Event immutability and reconstructability.

Storage technology SHALL remain replaceable.

8. Event Recovery

Recovery SHALL reconstruct constitutional behavior using Event history.

Recovery SHALL preserve:

Identity.

Relationships.

Architectural behavior.

Historical continuity.

Equivalent constitutional behavior.

9. Event Federation

Federated implementations SHALL exchange Events while preserving:

Origin.

Ordering.

Canonical identity.

Relationship integrity.

Historical attribution.

Events SHALL remain locally owned.

10. Event Verification

Verification SHALL demonstrate:

Immutability.

Historical completeness.

Correct ordering.

Projection equivalence.

Recovery correctness.

Equivalent constitutional Event history.

11. Event Pattern

Implementation SHALL optimize Event processing.

Implementation SHALL NOT optimize away Event preservation.

Operational efficiency SHALL remain subordinate to constitutional continuity.

12. Event Implementation Test

Event Implementation satisfies this specification when two independent implementations can reconstruct equivalent constitutional history solely from their preserved Event collections despite differing execution models, storage technologies, and projection strategies.

Conformance

An implementation conforms to this specification when Events remain immutable, reconstructable, historically continuous, operationally equivalent, and constitutionally authoritative independently of implementation technology.

Implementation Principle

Applications usually remember the present.

Organodynamics remembers every constitutional step that created the present.

Implementation therefore treats operational state as temporary,

while Events become the permanent operational expression of constitutional history.

Everything may be recalculated.

History may only be extended.

End of Specification OIS-004